



# Shadow Science

A Beginner STEM Journey Exploring Light  
& Shadows Through Indian Culture  
Planrs Academy



# Welcome to Shadow Science

Discover the magic of light and shadows through curiosity-driven experiments, animated stories, and hands-on activities rooted in Indian culture.



Understand how shadows form



Predict shadow sizes based on light position



Observe shadow changes through the day



# What Is a Shadow?

A shadow is a dark area formed when an object blocks light. Light always travels in straight lines, so when something gets in its way, it creates a shadow behind it!



A shadow forms when light cannot pass through an object



Light travels in straight lines from the source



Stand in the afternoon sun and see your own shadow!





# Light Sources Around Us

Light comes from many sources! Some are natural like the Sun and fire. Others are artificial like torches, tube lights, and diyas. Each creates shadows in its own special way.



Natural light: Sun and fire give us bright, warm light



Artificial light: Torch, tube light, and diya lamps



Diwali diyas create flickering shadows!





# Big Shadow, Small Shadow

Shadow size changes based on how close or far the light source is from an object. The sun's position in the sky also affects shadow length throughout the day.



Light close to object creates a big shadow



Light far from object creates a small shadow



Noon in Chennai: tiny shadow,  
Evening in Jaipur: long shadow

# Opaque, Translucent, Transparent Objects

Different materials affect how light passes through them, creating different types of shadows. Let's explore three categories!



**Opaque:** Blocks all light → dark shadow (steel tiffin box)



**Translucent:** Some light passes → faint shadow (butter paper)



**Transparent:** Most light passes → almost no shadow (glass of water)





# Shadows Change Through the Day

Watch how shadows move and change size as the sun travels across the sky! Punjab farmers have tracked time by shadow length for generations.



Morning: Sun low in east creates long shadows pointing west



Noon: Sun overhead makes short shadows directly below



Evening: Sun low in west creates long shadows pointing east





# Culture Connect: Shadow Puppetry

India's shadow puppet traditions blend art and science! Tholpavakoothu from Kerala and Togalu Gombeyaata from Karnataka use opaque leather puppets, translucent cloth screens, and oil lamps to create magical moving shadows.







# Discover, Play, and Learn with Shadows!

Become a Shadow Scientist and explore the magical world of light and shadows all around you! Join Planrs Academy's Shadow Science course today.





# Thank You for Exploring Shadow Science!

Keep observing shadows every day and become a true Shadow Scientist! Visit Planrs Academy to register for the full course and continue your STEM journey.

